

# Schools Turnover in Chile (1992-2012)

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*An analysis of the primary and secondary level of Chilean educational system is presented through the study of the schools that enter and leave it, with special emphasis on those schools that receive subsidies of the state. The most relevant results found show that the schools that close, regardless of their administrative dependence (funding), have on average more vulnerable students. Second, the schools that close have lower scores in the standardized tests (SIMCE and PSU), a result that is relativized by correcting the level of vulnerability of the establishment, showing that only those out of the less vulnerable quintiles have statistically worse results than incumbents (those schools that do not close). Third, the evidence suggests that the monitor and control mechanisms carried out by the state have effects on the probability of closing a school, a situation that, despite not meaning causality is evidence that support the inspection system. Fourth, schools that close show a lower average monthly paid compared to incumbents (are cheaper). Relevant fact by virtue of the findings, since this implies that higher paid do not necessarily translate into better quality. Fifth, the concentration of schools at the municipal level has effects on total turnover, showing that in those with a higher Herfindahl-Hirschman Index (HHI), after correcting for demographic and socioeconomic characteristics, present higher probability of school entrance and more important, greater number of total closures.*

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## I. Introduction

An analysis is presented that pursues two objectives: first, to identify the main differences in key variables between the schools that close and the incumbents during the last 20 years. And second, to contribute to the identification of the determinants of the opening of approximately 5000 and closure of 3000 schools in the Chilean educational system in its basic and middle levels since 1992. The research includes a multifactorial analysis of territorial variables, regulation and sectoral legislation pertaining to the period, characterization of the commune (through socioeconomic variables, expenditure, structure and composition of the educational system), characteristics at the level of schools (socioeconomic, type of teaching provided and the administrative dependence of the schools) and characteristics of the students (socioeconomic and their performances in the standardized tests).

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The transformation of the system in terms of its composition by administrative dependencies will be analyzed, which has been studied by many authors, among others by Corvalán et al. (2008) and Elacqua (2009), however, and despite the previous analysis, it is possible to perform an additional geographical analysis that helps to show the dissimilar evolution of the composition of dependencies between communes and regions. It will also show a phenomenon of inertia in the composition when it is extreme, that is, when within a commune there is a predominant type of administrative dependence<sup>1</sup>. The work is structured as follows, in the first part there is evidence of total turnover of schools, their temporal evolution, and a territorial disaggregation. In the second part, the outgoing schools are characterized and separated from the incumbents by means of performance variables in standardized tests, type of financing, territorial disaggregation and socioeconomic level, in the search of patterns that determine significant differences between both. One-third uses variables at the municipal level to account for socioeconomic patterns, industry concentration and performance on standardized tests that help explain the different levels of turnover found. In the fourth and final part, stylized facts previously found are estimated using econometric models. It is worth mentioning that the object of study of the research is the complete educational system, but without prejudice to the above, it will be analyzed with special interest the charters schools (subsidized by state), the potential gain of their supporters<sup>2</sup> and the volume reached by the rapid growth since the nineties.

## II. Stylized Facts

In 1992 there were approximately 10000 schools, while in 2012 there were about 12000. Considering the entrances and closures the total of schools between 1992 and 2012 was of 15000. In 1992, of the total of schools, 10% worked in the private modality, 27% as charter schools and a majority of 63% worked under the municipal administration. In 2012 the composition was very different with 5.7% under private administration, 49% in charter and 45.3% in municipal<sup>3</sup>. These photographs taken at two points of time do not account for the entry and closure between periods, and the revised literature only tangentially addresses the dynamics and the determination of factors that cause the closure and entry of schools<sup>4</sup>, without this being the center of analysis.

<sup>1</sup>Some possible explanations for this phenomenon can be attributed to the role played by the preferences of the proxies, which may be determined by your salary or creed. Those are heterogeneously distributed within the regions, and more homogeneously within the communes

<sup>2</sup>According to figures in Corvalán et al. (2010), 71% of charter schools are for profit.

<sup>3</sup>For historical figures see Aedo and Sapelli (2001) and Hamuy (1960) cited by Elacqua in "Brief history of educational reforms in Chile (1813-present)".

<sup>4</sup>It is worth mentioning the undergraduate thesis elaborated by De Iruarizaga (2008), who deals with the dynamics of the closure of charter schools that provide basic education.

#### A. Entrances and closures, total and net turnover

Total turnover is defined as the sum of the entrances and closures of schools. In industrial organization, total turnover is also known as total reallocation of signatures. In the case of usual goods and services, it is interpreted as a measure of both dynamism of the industry and efficiency, under the assumption that at higher levels of this, a creative destruction occurs with efficiency gains (the least productive ones are displaced out of the market by more productive firms<sup>5</sup>). However, it is not clear that this interpretation applies in the case of the education system, for at least two reasons: in the first place, it is not obvious that the closure of a school is caused by its low productivity, since the outgoing schools can simply be the less profitable schools and not necessarily, those that deliver the worst quality of education after controlling for socioeconomic factors. Secondly, unlike a usual good such as a pencil, a food or clothing, the closure of schools could be associated with economic and social costs for the students that could well surpass the eventual benefits of the reassignment. When looking at the figures the total turnover<sup>6</sup> The type of school with most dynamism are charters, representing 65% of total entries and 33% of the total of closures of the period. Municipal education, meanwhile, leads the destruction with 50% of the total and, registers the second lowest number of entries with 19% of the total. The private sector has the lowest number of closures (17% of the total).

These results are not surprising, since this composition is a consequence, deliberate or not, of the structure of the educational system and its economic and legal incentives in the period analyzed. In practice, the increase in coverage has been privileged, based on the expansion of charter schools, a fact that has been rescued in the analysis of educational policies by both Cox (2003) and UNESCO (2004). In particular, Cox argues that the expansion of this type of schools took place at the expense of the municipal sector: "The new sector of charter schools emerged during the 80s, would have competed not in terms of increases in their productivity, but selecting the best students, and thus producing a *discreet* of the so-called fiscal education". This, in spite of the fact that the evidence is not conclusive in favor of a better performance of subsidized students over municipal ones (review among others Lara et al. (2011)<sup>7</sup> and Paredes and Drago (2011)<sup>8</sup>). At the same time, the public offer was strongly limited by the capacity of the munic-

<sup>5</sup>Griliches and Regev (1995), who show for Israeli firms that the increases in productivity of the industry come from internal improvements of incumbent firms, and not from increases in productivity as a result of the reallocation of signatures, that is, and for this case, the *creative destruction* defined by Schumpeter would not be happening.

<sup>6</sup>Total turnover is calculated using the directory of schools made available by the Ministry of Education since 1992, and supplemented with information at the level of schools and communes. In the present methodology all anomalous records are identified and excluded from the calculated turnover.

<sup>7</sup>They show by means of matching models, that the differences in favor of charter schools are small, and sometimes not statistically significant, of the order of 4 to 6% of the standard deviation of the test.

<sup>8</sup>They perform a meta-analysis based on to 17 articles concluding that there is a difference of 4 points in favor of the charter schools by over the municipal ones, around one tenth of the standard deviation presented by the test used by them (SIMCE of Mathematics of 4th basic of 2008).

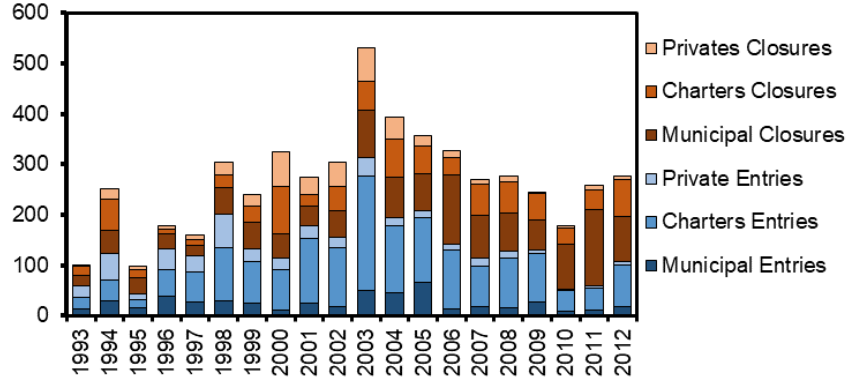


Figure 1. : Total turnover by type of school (funding)

*Note:* (E) Entries, (S) Closures.

*Source:* Based on MINEDUC data.

ipalities to expand in the sectors with the highest population growth. The figures 2 and 3 present evidence of total turnover, creation and destruction of schools disaggregated by administrative dependence, first of all, there are important differences of turnover according to educational dependency, being the full-paying students the most volatile with destruction rates exceeding 9% (2003), well above the maximum of charter students of 3.5% (2000), and 2.8% (2011) of the municipal. On the creation side, the order is as follows: full-paying students (8% in 1998), charter (7.4% in 2003) and municipal (1.1% in 2005). Second, turnover in charter schools begins to take off only after 1995, with rates that exceed 3% for more than 10 years. Finally, the average turnover rates for the 20 years are 6.3% for private, 4.5% for charter and 1.6% for municipal schools.

Although not be an industry of goods with an industry that provides services, and understanding that the Chilean educational system works as a market with competitors for the same demand, the rates of entry and closure are similar to those obtained by Külzer and Benavente (2008)<sup>9</sup> For a sample of companies from different sectors. The average creation and destruction rates between the years 1998 and 2006 for medium-sized companies were 2.2% and 1.0% respectively. They also find that the rates grow inversely proportional to the size of the companies, for example, micro-enterprises have creation rates of around 15% and destruction of 12%. In the quasi-market of education, the results indicate that municipal schools have creation rates of 0.4% and destruction of 1.2%, lower than those exhibited by charter students of 2.9% and 1, 5% respectively. However, the

<sup>9</sup>In this study they find that those sectors that present greater economies of scale tend to have a lower entry rate. On the other hand, the variables of economic environment (GDP and unemployment) affect destruction rather than the creation of new companies. They find a strong persistence effect, where the closure rate of companies observed in the past consistently impacts the current creation rate.

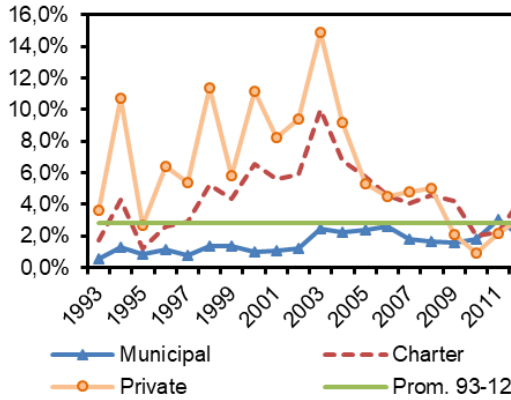


Figure 2. : Relative turnover by type of school (funding)

Note:  $Turnover_t = entries_t + closures_t$

Source: Based on MINEDUC data.

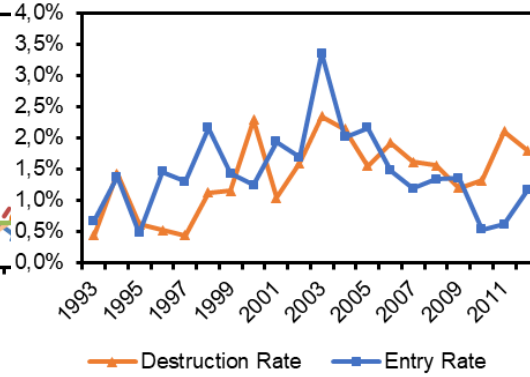


Figure 3. : Rate of creation and destruction of schools

Note:  $Creation = entries_t / stock_t$

$Destruction = closures_t / stock_{t-1}$

Source: Based on MINEDUC data.

average creation and destruction rates for the period 1993 to 2012 for all dependencies is 1.5% and 1.4% respectively. These rates represent the income of 2.160 schools and the closure of 2.065 at the country level<sup>10</sup>

The figure 4 shows the net turnover in which the above mentioned phenomenon can be appreciated more clearly, and another one is evidenced in particular for the private schools. Where the data suggest a greater sensitivity to the macroeconomic environment of these schools, evidencing a decline in the stock that is absorbed in part by revenues from municipal schools, and mainly by charter students (this fact is later taken up in the econometric analysis). It is precisely the closures that occur due to the instability of the economic cycle that are tremendously undesirable. Both in terms of efficiency in the use of resources, and mainly in the safety and quality of the educational process of the students. Moreover, and as will be shown in the following sections, if finally the schools with the most vulnerable students are the ones that close.

The absorption of the registration of the municipal ones is carried out mainly by the charter ones, evident situation for the year 1998, 2003 and 2006 where the paths are partially opposed. Another striking fact is the closure of municipal schools in the years where secondary student mobilizations took place, in 2006 and 2011. They closed more than 250 schools throughout the country. Releasing a demand that was absorbed mainly by new charter students, which grow in 200

<sup>10</sup>They differ from the previously presented numbers since now all the educational schools that give exclusively are excluded: kindergartens, adults, education in prison and special (or some combination of the above), those schools that enter and leave the following year, and those that appear as closures or entries only by changing the role of the database, but maintain their functioning under another role. All the presented analysis, with the exception of the figures indicated, uses this subset of data.

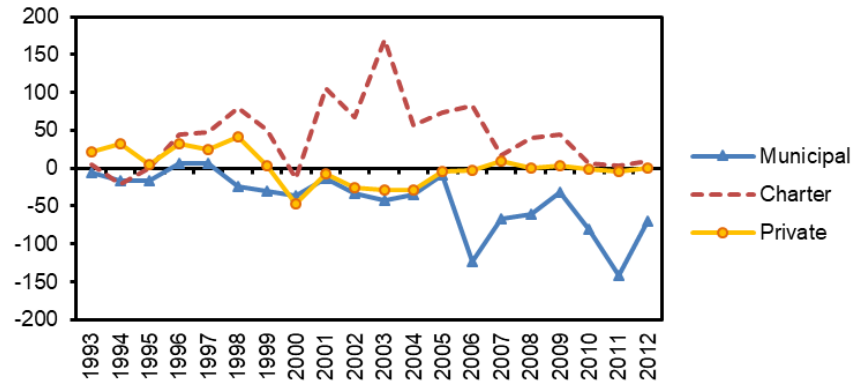


Figure 4. : Net turnover by type of school (funding)

Source: Based on MINEDUC data.

schools in the years after 2006.

#### B. Legal framework

The legal and regulatory aspects will be considered as milestones that change the incentives for entry and closure of schools, as well as the conversion between administrative units. The relevant milestones of the period under consideration are: the enactment of the shared financing law of September 1993 (amendment to DFL N°5 of 1992), the existence of support programs (incentive) for schools with the worst (best) results such as the P-900 of 1990, the Enlaces program of 1992, the basic MECE between 1992 and 1996, the Pedagogical Excellence Assignment (AEP) program of 2002, the full school day (JEC) and its gradual incorporation since 1997, the official recognition of nursery-level education in Law 18,962 in April 2003, the entry into force of the preferential school subsidy (SEP) in 2008, and the successive increases to the regular monthly subsidy per student throughout the period analyzed ( 1995, 1998, 2000, 2002, 2006, 2008 and 2012<sup>11</sup>). Some of these educational policies and programs<sup>12</sup> Have dynamic consequences beyond the scope for which they were designed, since they change the incentive scheme of both municipalities and holders and potential investors. Thus, for example, increases in the regular and preferential school subsidy per student displace both the income and the administrators' costs curve, but not necessarily in the same magnitude, and there are no mechanisms to ensure that the largest resources transferred through grants are destined to improvements in the product and/or

<sup>11</sup>Increases to the regular subsidy fixed by laws Nos. 19,398, 19,662, 19,715, 19,808 and 20,637, article 7° law N°19.933 and decrees Nos. 37, 40, 58, 166 and 223 d the MINEDUC.

<sup>12</sup>See details of the programs in UNESCO (2004), Elacqua (2011) and Cox (1997, ?, ?).

educational process of students<sup>13</sup>. Thus, if the movements are parallel and in equal magnitude, we are in a balance where the holder rents the same as before, but at a higher level of quality. But there may be the worst scenario where the largest subsidy is transferred exclusively to the profits of the administrators of the schools.

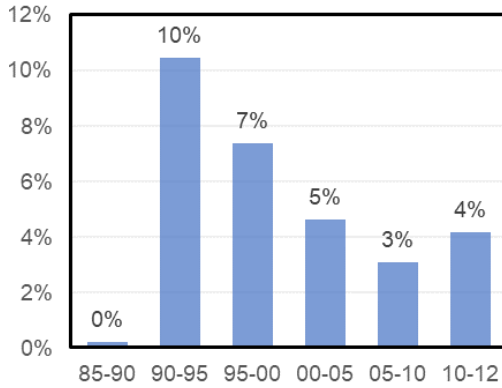


Figure 5. : Subsidy Amount (annual growth rate)

Source: Cox (1997), Elacqua (2011) and MINEDUC (2012).

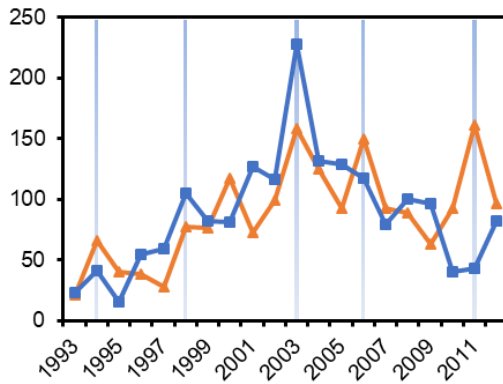


Figure 6. : Charter schools entries and total closures of private and municipal schools

Source: Based on MINEDUC data.

The inflows and outflows suggest that the regulatory changes linked to the subsidies of both the 1980s and the 1990s have had effects attributable to them, although it is not possible to demonstrate causality, they have arisen in the years immediately following entry of both new holders, and reconversion of private schools. It can be assumed on the latter, that the increase of the subsidy per student was at least attractive enough to leave them in the same curve isobenefits that before<sup>14</sup>. Following this logic, the years 1994, 1998, 2003, 2006 and 2011 generate closures from bulky municipal schools, approximately twice as much as the previous year for all cases. Same situation occurs in the 90s for full-paying students. At least in the visual inspection (see figures 5 and 6), there seems to be a relationship between the inflows of charter students and the modifications to the amount of the subsidy. This side effect not intended by the original public policy, if effective, generates a depredation of municipal education that translates into a deepening of social segregation (Hsieh and Urquiola 2003), even more so

<sup>13</sup>Furthermore, it is worth mentioning that there are improvements in the quality of the educational process that are not perfectly measurable as improvements in infrastructure or areas of knowledge outside the measurement of SIMCE or PSU, such as physical education and the arts.

<sup>14</sup>This turnover of schools reconverted in the periods analyzed raises additional questions for example, if there are statistically significant differences in measurable academic performance after the change of dependency, a phenomenon that will not be addressed here.

if we know that charter schools have entrants in average copayments higher than outgoing ones. That is to say, within the group of schools that operate under the modality of shared financing, the schools that have entered the system charge on average monthly payments greater than those charged by those who have closed. This fact is transversal between regions of the country.

Table 1—: Average Monthly Payment 2004-2011

| Situation | Average | Standard Error | 95% C.I       |
|-----------|---------|----------------|---------------|
| Entrances | 17.051  | 207            | 16.646-17.456 |
| Closures  | 9.666   | 634            | 8.423-10.909  |

*Source:* Based on MINEDUC data (Obs.: 5.465).

A second example is the entry into force of the shared financing system<sup>15</sup>, Which already in 1996 (only three years after its promulgation) achieved that 65% of students belonging to charter schools were under this modality, almost a quarter of the total schools of that year (Aedo and Sapelli 2001). Adoption at the municipal level was lower in the beginning, mainly because they can only receive shared funding at their average level, unlike charter students who can do it both at the primary and secondary level, this situation generated less adoption of the shared financing between the municipal schools, which have never exceeded 25% of the total enrollment of students. While the charter reaches 80% of enrollment today. In 1994, the highest number of closures of private students was recorded, mainly those that offer basic and kindergarten education. Meanwhile, in the charter schools enter about 40 schools, the highest number of the first five-year period of the nineties, comparable with the entry rates that would be maintained from 1998 onwards, except for the years 2009 to 2011 that followed the financial crisis . This change in the composition of schools towards one with a greater number of charter students occurs in two ways: openings of schools not related to those that close, and those that were private full-paying students and converted to subsidized. It is possible to verify that one of every five openings in this period corresponds to the reconversion of administrative dependencies. The Full School Day (JEC)<sup>16</sup> On the other hand, it had a gradual adoption with two important milestones: first , the program started in 1998, during which 686 schools were incorporated. Second, the adoption deadlines established as mandatory for the municipal sector and the charter schools, were the beginning of the 2002 school year, and then extended at the beginning of 2007 for these schools and 2010 for subsidized non-vulnerable students. Potentially these maturities could generate closures, but in practice the situation seems not to have represented an active

<sup>15</sup>For more information, review law No. 19.247 enacted on September 9, 1993 and Vial (1998).

<sup>16</sup>Law No. 19.532 of November 17, 1997, and subsequently amended in 2004 and 2011 by law No. 19.979.



restriction for the holders and municipalities, and the closures observed in these milestones are not significantly different from those observed in other years.

### C. Regional and Comunal differences

There are large differences in flows between regions, and within them between their capital and the rest of the cities that make up the region. For example, the region of *Araucanía* had a growth of charter students below the national average that was approximately 2.1% in the period, adding about 4 schools per year at a growth rate close to 1% . On the other hand, the region of *Arica and Parinacota* added in the same period 3 schools per year at a rate of approximately 8.1%, in other words, in 20 years more than quadrupled their allocation of charter schools. For regions VI to X and XIV (center-south of the country) total turnover of the school system occurs mainly in medium-sized cities and smaller (less than 15000 inhabitants), which on average represent 80% of the total turnover of the mentioned regions.

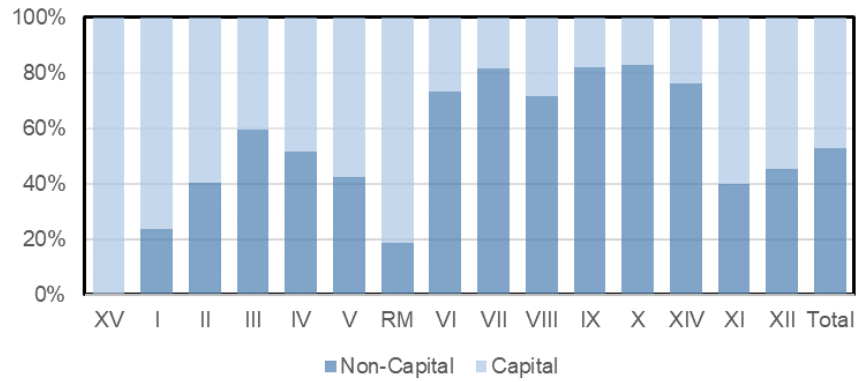


Figure 7. : Total turnover by region and capitals

Source: Based on MINEDUC data.

This is consistent with the fact that the regional population is not as concentrated in the regional capital as it is in the Metropolitan Region, where Santiago agglomerates 94% of the school-age population of the region projected for the year 2012, in comparison with O'Higgins and Maule where that figure reaches only 32%. This phenomenon is repeated in Valparaíso, whose regional capital represents 58% of total turnover and has 50% of the school-age population in the region. In Biobío, meanwhile, 48% of the school-age population lives in the Gran Concepción, but only 29% of total turnover is explained by the regional capital. These disparate results show that the scale of the cities and the size of the market have effects on the level of inputs and outputs of the industry, but the sign

is far from clear. Despite this, it seems that in small cities -even more in towns or villages- families have limited options and low competition that generates a significant inertia<sup>17</sup> Of the 346 communes only seven communes have a turnover greater than 5% of their stock on average in the 20 years.

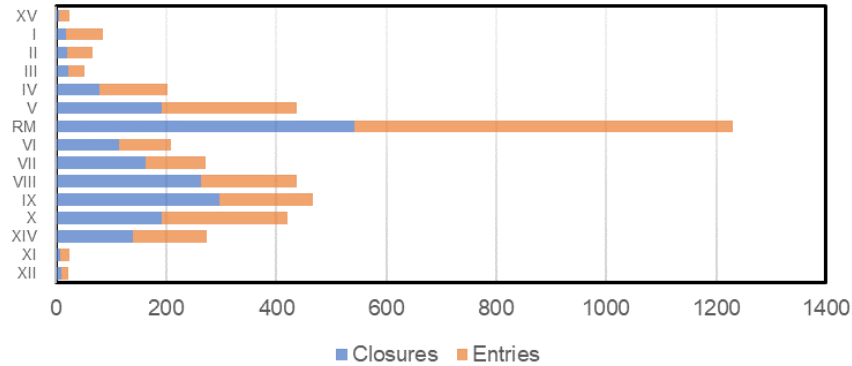


Figure 8. : Total number of entrances and closures by region

Source: Based on MINEDUC data.

COMMUNES OF THE METROPOLITAN REGION. — As happens between regions and regional capitals, within the Metropolitan Area of Santiago (composed of 34 communes) the total turnover is very heterogeneous, at one extreme we have communes with many outlets and entrances such as La Florida and Puente Alto with 65 entrances each, and on the other, Lo Prado and Pedro Aguirre Cerda with 4 and 5 respectively. Possible determinants of this heterogeneity, may be the differences in labor income among the inhabitants of the communes, and a particularly good approximation in that line is the School Vulnerability Index (IVE-SINAE), occupied by the MINEDUC to identify and prioritize the students and their schools according to their vulnerability or risk. It is appreciated that there is a statistically significant difference between the schools that close and those that remain open according to the percentage of vulnerable students who attend them. This difference is accentuated in the case of charter students, where on average their enrollments are 62% of students categorized as vulnerable, raising that figure to 72% in the group of schools that close during the period analyzed. For municipal students, the same thing happens: the average number of vulnerable students in schools that close is 87%, nine percent points more than the average for municipal schools that do not close.

<sup>17</sup>See Elacqua et al. (2012) which shows evidence on the feasibility of closing certain schools badly evaluated when the set of next options is reduced in the best of cases, but null.

Table 2—: Percentage of enrollment with vulnerable students according to type of school (funding) and situation, Santiago

| Year  | Municipal |       | Charter Schools |       | Total |       |
|-------|-----------|-------|-----------------|-------|-------|-------|
|       | Open      | Close | Open            | Close | Open  | Close |
| 2007  | 82,7      | 89,1  | 65,4            | 74    | 71,8  | 76,8  |
| 2008  | 89        | 93,6  | 71,7            | 79,4  | 78    | 82,3  |
| 2009  | 77,5      | 85,4  | 62,4            | 70,3  | 67,8  | 74,2  |
| 2010  | 71,8      | 78,4  | 57,8            | 65,6  | 62,8  | 67,3  |
| 2011  | 74,1      | 84,3  | 59,5            | 67,4  | 64,6  | 68,6  |
| 2012  | 75        | 87,3  | 60,7            | 73    | 65,7  | 74,6  |
| Total | 78,3      | 87,4  | 62,8            | 72,2  | 68,4  | 74,9  |

Source: Based on MINEDUC data (Obs.: 8.089).

However, despite the phenomenon found above, it is observed that the greater turnover occurs in less vulnerable communes, this may be due to the administrative dependency composition described above. In more vulnerable communes there are fewer charters schools, which are precisely those with greater degrees of total turnover. So it is not surprising that on average less vulnerable communes have higher rates of destruction and creation. The relevant finding here is undoubtedly that the destruction occurs in more vulnerable students.

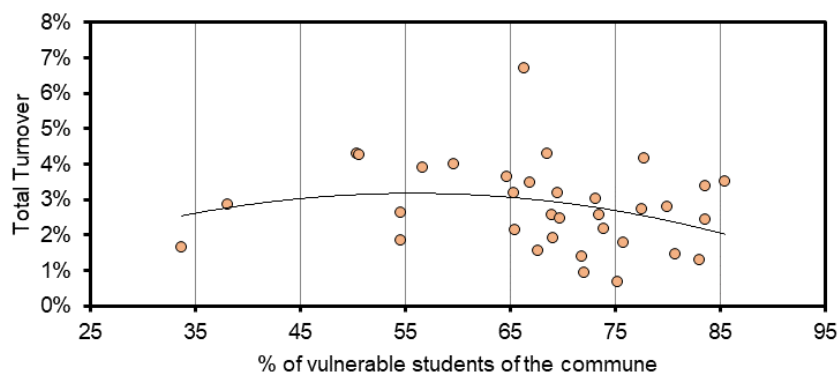


Figure 9. : Relation of total turnover and socioeconomic background, Santiago

Source: Based on MINEDUC data.

These differences are transversal to the region, and it is possible to observe the same phenomenon in the rest of the country as well. The cloud of points shown in the figure 9 shows the existing relationship for Santiago between the Vulnerability Index (IVE-SINAE) calculated by commune and the total turnover at the level

for the whole period.

PRIMARY AND SECONDARY STAGES. — It can be seen in the figure 10 that the type of education given by the schools, both in those that leave and those that enter, is mainly some combination of basic with kindergartens and/or half humanistic scientific education. The schools that exclusively offer special education for adults or in prisons have been excluded from the analysis because they correspond to an educational demand different from that analyzed.

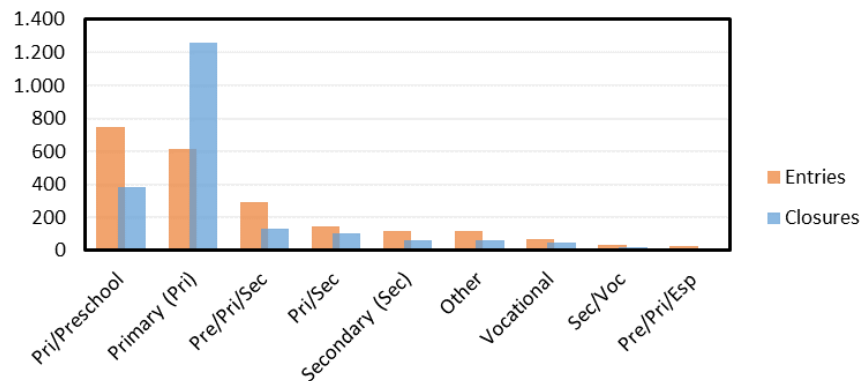


Figure 10. : Total number of entrances and closures by stage of education

Source: Based on MINEDUC data.

Of the total number of closures for schools offering basic education exclusively, approximately 950 (76%) correspond to closures of municipal schools, mainly in the regions of Biobío, Araucanía and Maule (54% of the total). On the side of the entries, 900 (67%) correspond to income of charter students, mainly schools that provide elementary and kindergarten or basic education exclusively. Of these, 52% occurs in only three regions: Metropolitan, Los Lagos and La Araucanía. The year with the highest income of basic schools was in 2003, coinciding with the enactment in April of Law 18,962, which grants, among other things, official recognition to schools that teach kindergarten. Fact that possibly explains to a large extent the income of 47 schools throughout the country under the basic modality plus kindergarten education, 21 of them only in the Metropolitan Region, all of them charter schools.

HERFINDAHL AND HIRSCHMAN INDEX (HHI). — The Herfindahl and Hirschman index is calculated by commune and year for the study period, which corresponds to a measure of economic concentration in a market. The higher the index, the more concentrated and less competitive the market. The concentration of the

industry seems to have important effects on total turnover, so very concentrated communes (rural municipalities with only one or two schools) show little or no turnover. However, the phenomenon is also extendable to urban districts such as Freirina, Huasco, Caldera and Mejillones, which, on average, have an IHH above 2500 (considered as an indicator of a highly concentrated industry) with very low total turnover, of the order of 1% of the total stock.

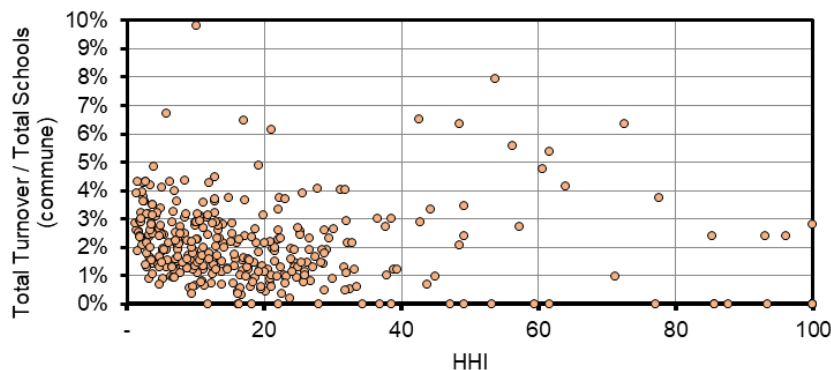


Figure 11. : Relationship between turnover and market concentration

Source: Based on MINEDUC data.

On the other hand, the communes where there is greater competition (less monopoly power) are in decreasing order Viña del Mar (1180), Florida (1380), Valparaíso (1400), Puente Alto (1530) and Santiago (1534). Being also the most populous communes and of greater population growth of the country between the years 2002-2012. However, due to the large volume of schools they have, the total turnover rates do not exceed 4%, being that on average they exceed the 40 schools created during the period.

#### D. Variables of quality of teaching

**RATE OF STUDENTS PER TEACHER.** — The ratio of students to teachers is an indirect indicator of the quality of the educational process as it is an approximation of the time allocated by each teacher to the training process of these, as the reason grows, less time the teacher allocates to each student. For the current financial year, this indicator is not entirely reliable, since the schools that are close to closing are steadily reducing their enrollments in previous years, at a different speed than the number of teachers. Thus the reason can lead to deception since in the years prior to the closure will have fewer students per teacher, a situation that probably is not due to a recruitment of new professionals (changes in the denominator). This deterioration effect prior to closure, called as *shadow of death* by Griliches

and Regev, can also affect other quality variables such as the standardized SIMCE and PSU tests.

Table 3—: Rate of students per teacher, by year and closure

| Year | Municipal |       | Charter School |       | Total |       |
|------|-----------|-------|----------------|-------|-------|-------|
|      | Open      | Close | Open           | Close | Open  | Close |
| 2004 | 20,7      | 14,9  | 24,9           | 19,9  | 21,8  | 16,2  |
| 2005 | 19,7      | 13,4  | 23,8           | 18,4  | 20,8  | 15    |
| 2006 | 19,4      | 12,7  | 23,8           | 18,4  | 21    | 14,9  |
| 2007 | 18,9      | 13,3  | 23,8           | 18,3  | 20,8  | 15,2  |
| 2008 | 18,2      | 12,2  | 23,4           | 16,9  | 20,4  | 13,9  |
| 2009 | 17,1      | 11,1  | 22,6           | 16,8  | 19,5  | 13,2  |
| 2010 | 16,5      | 10,6  | 22,4           | 16,1  | 19,2  | 12,8  |
| 2011 | 16,1      | 10,3  | 21,7           | 17,2  | 18,9  | 15,2  |

*Source:* Based on MINEDUC data (Obs.: 67.116).

STANDARDIZED TESTS: SIMCE AND PSU. — The analysis made compares the results obtained by the schools that close with those that remain open during the entire analysis period, without controlling for economic or sociodemographic differences. The exercise is carried out considering all available information of the SIMCE of mathematics for primary and secondary levels of the whole country, between the years 1998 and 2011. The comparison of means is carried out naturally between schools that have rendered the measurement by level, that is, for primary the sample is composed of municipal, charter and private schools. This allows to work with a larger sample and therefore more robust confidence intervals. The results show consistently, indistinct of the level in which the SIMCE is carried out, that the schools that close obtain on average lower scores than their peers still open. For the tests of mathematics in primary level the differences of the score obtained, measured as a percentage above or below the communal average, is 4.3% less for the case of charter schools that close. And 2.2% more than the communal average for the case of those who remain open. The situation is not different for the municipal ones, only that for these in both scenarios (closing and not closing) the communal SIMCE obtained is below the communal average, but the difference is still statistically significant in favor of those who do not close, which they are 2.5% less than the communal average.

The phenomenon is stable both in time and between levels of education, and even by area of knowledge, since despite being less pronounced the difference in language, the difference in means found is statistically significant. An additional stylized fact that is worth mentioning, relates to the apparent exacerbation of the gap at higher levels (see figures 14 and 15), where the difference between charter

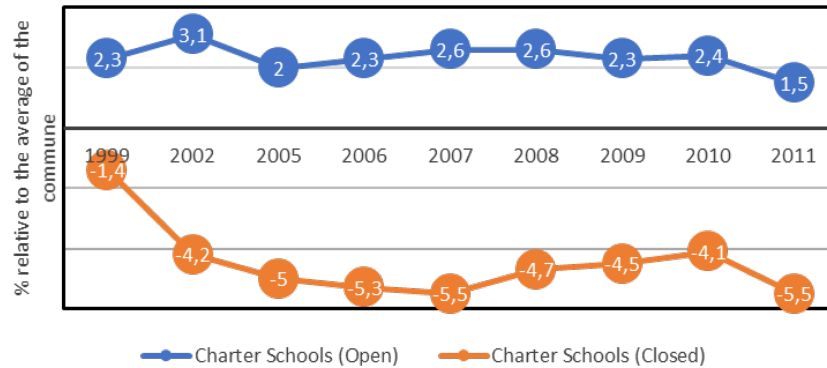


Figure 12. : Math SIMCE 4th. Grade, differences with the communal average by closing situation

Source: Based on MINEDUC data.

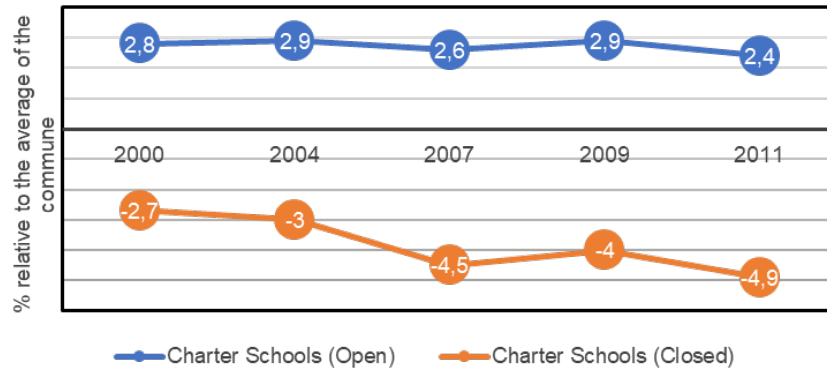


Figure 13. : Math SIMCE 8th. Grade, differences with the communal average by closing situation

Source: Based on MINEDUC data.

students that close and those that do not for the SIMCE of second means and the PSU is significant and of approximately 10%<sup>18</sup>.

A complementary analysis now controlling for vulnerability of schools, measured in quintiles of the IVE-SINAE by region and year, allows to observe existing differences between outgoing, incoming and non-mobile schools in the period<sup>19</sup>. The figure 16 shows the average (horizontal black bar) and its 95% confidence

<sup>18</sup>What is found does not seem to depend on the territorial unit of analysis, since similar results are found when carrying out the previous exercise in the different regions, where in particular the gaps are greater for the Great Concepción.

<sup>19</sup>Exercise carried out for SIMCE of mathematics of the year 2008, which has more observations than other years product of natural attrition when approaching the end of the panel. To preserve a symmetric study window, they are considered schools that enter and leave between 2003 and 2013.

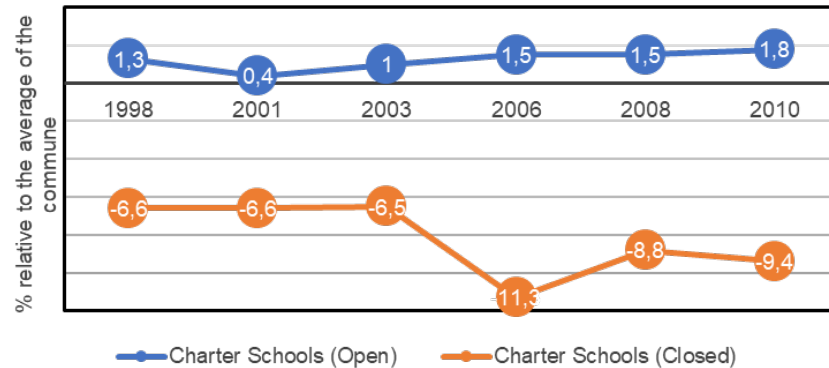


Figure 14. : Math SIMCE 10th. Grade, differences with the communal average by closing situation

Source: Based on MINEDUC data.

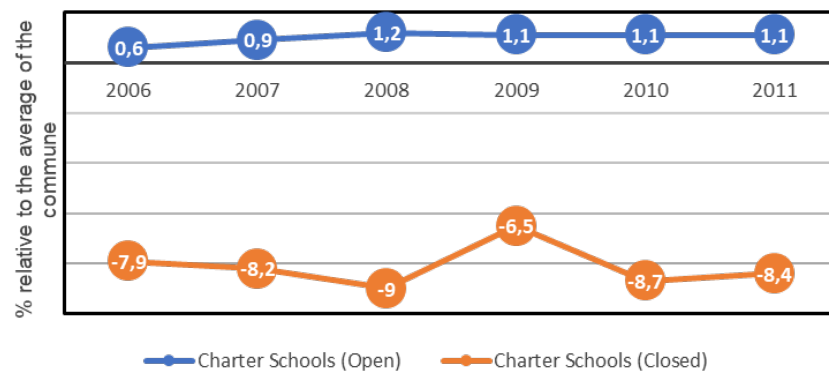


Figure 15. : Math PSU 12th. Grade, differences with the communal average by closing situation

Source: Based on MINEDUC data.

interval (rectangle). The differences of calculated means are not statistically significant for the first three quintiles, there being processes of *schumpeterian improvements* only in the two quintiles of higher income (schools that enter have better results on average than those that leave the system).

This suggests not only the existence of a great heterogeneity in the scores according to their vulnerability, but also that the destruction and creation of new schools have at least questionable effects in the improvement of the educational system, since they do not generate transversal quality increases between different socioeconomic groups, and if they existed, seemed to be limited to only the wealthiest quintiles of the country.



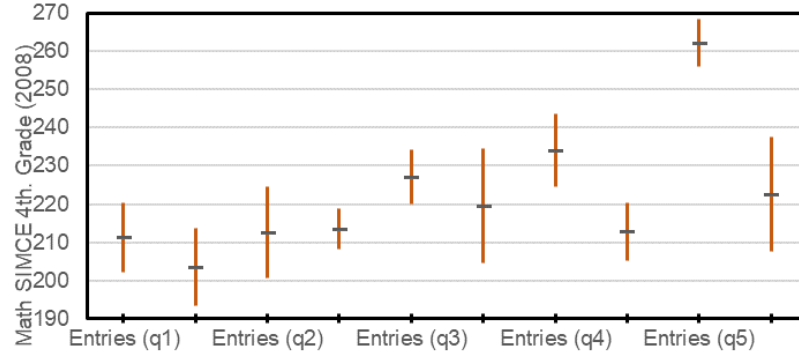


Figure 16. : Mean and C.I by socioeconomic groups, Math SIMCE 4th. Grade (2008)

*Source:* Based on MINEDUC data.

Table 4—: Total turnover by socioeconomic groups (IVE)

| Turnover | Q1 | Q2  | Q3  | Q4  | Q5  | Total |
|----------|----|-----|-----|-----|-----|-------|
| Entries  | 13 | 48  | 83  | 123 | 261 | 528   |
| Closures | 40 | 257 | 32  | 54  | 44  | 427   |
| Total    | 53 | 305 | 115 | 177 | 305 | 955   |

*Source:* Based on MINEDUC data.

### III. Econometric Analysis

The stylized facts presented above are complemented by an econometric analysis at the level of students, schools and municipalities. First, a duration model is estimated to identify the determinants of school closures. Second, a linear multi-level model (hierarchical or HLM) is estimated to verify if there are demonstrable differences in the results of the SIMCE test between the schools that remain active and those that close, controlling for heterogeneity at the student level and schools. Third, a multilevel model with longitudinal data to explain changes in the enrollment of schools. Finally, principal determinants of closures at the commune level are estimated using a linear multilevel model.

#### A. Closure Probability

The model chosen to estimate the probability of closure of schools is a model of duration considering time as a discrete unit (Beck et al. (1998)). This is because the closure can be understood as episodes, where the milestone is marked by the

end of the school year, and the consequent decision of the administrator or holder to continue or close the school next year<sup>20</sup> The estimation of this model makes use of the properties of the sample verisimilitude, which can be rewritten as the likelihood of multiple regression with dichotomous dependent variable (Allison (1984), Jenkins (1995)). The formula for verisimilitude is given by:

$$(1) \quad \mathcal{L} = \prod_i^n \{h(t_i)S(t_{i-1})\}^{y_{ti}} S(t_{i-1})^{1-y_{ti}} \\ = \prod_i^n \left[ h(t_i) \prod_{i=1}^{t-1} (1 - h(t_i)) \right]^{y_{ti}} \left[ \prod_{i=1}^t (1 - h(t_i)) \right]^{y_{ti}}$$

Now, leaving the risk function  $h$  conditional to the regressors used in the model. This can be written as:

$$h(t_i, X) = Pr(T = t_i | T \geq t_i, X)$$

The probability of closure is given by  $Pr(Y_i = 1) = \lambda_i$ , and the probability of staying open by  $Pr(Y_i = 0) = 1 - \lambda_i$ . Then, the probability is a function of the regressors, being expressed as:

$$\lambda_{it} = \beta_0 + \beta_1 X_{1i} + \dots + \beta_k X_{ki}$$

The choice of the distribution function is the complementary log-log because it has statistical virtues in asymmetric distributions of the dependent variable, a situation that describes the present database (see figure ref fig:tasas). Finally, the probability of closure is expressed as follows:

$$(2) \quad \log[-\log(1 - \lambda_i)] = \beta_0 + \beta_1 X_{1i} + \dots + \beta_k X_{ki} \\ \lambda_i = 1 - e^{-e^{X_i \beta}}$$

For the choice of estimation regressors, the literature specialized in the analysis of entry and closure of companies (following Külzer and Benavente (2008) and Caves (1998)) suggest from theoretical models, and subject to availability and feasibility of constructing or obtaining the relevant variables, a set of indicators that can be summarized in three groups <sup>21</sup>:

- Scale factors and idiosyncratic characteristics of the firms, the first ones associated with sunk costs and barriers to entry. Measured in the present

<sup>20</sup>Estimated, as an alternative modeling approach, multi-level panel data models of dichotomous dependent variable, controlling for heterogeneity within each school. These models are less theoretically adequate for the type of data that is available.

<sup>21</sup>Technological regimes are considered in the literature as an additional factor, however, they do not seem to have an important role for the case analyzed.

work as the maximum occupational load of students who own a school in the period considered, in relation to the capacities of the other schools at the municipal level. Transforming this relative burden into an indicator with five levels, where one is for those schools that are in the 20% of the lowest installed capacity in the commune, and five for schools that are in the top 20%. The second set of indicators, meanwhile, is composed of establishment variables such as: administrative dependence, type of education (confessional or secular) and school performance (SIMCE results and magnitude of repetition)

- Concentration and market growth, the first measured in the present work by the concentration (Herfindahl-Hirschman Index) at the municipal level. And the second by the growth rates of the school-age population, at the municipal level as well
- Global economic cycle, as an exogenous cause of a greater or lesser dynamism in the creation or destruction of signatures. Variables are incorporated in the present work that capture changes in the economic cycle in the period considered, such as the GDP growth rate and the national unemployment rate.

Mixed evidence is found that supports some results presented previously in the work. In particular, it can be seen that the relationship between SIMCE and closures is weak after controlling for characteristics of the market and schools. In the table 5 the final model is presented, it can be seen that the coefficient associated with SIMCE has a negative effect on the probability of closure. That is to say, that in front of increases of score a decrease of the probability of closure is observed. To illustrate the magnitude of the effect, if an establishment increases by 50 points its average in the SIMCE (approximately one standard deviation) it reduces the probability of closing by 0.31 percent points. Considering that the average annual closure of schools in the last 20 years was 1.41%, this increase in SIMCE would make the probability fall to 1.01%. It is worth noting that, in educational outcomes, a change of one standard deviation is much higher than the effect of any policy.

On the other hand, for the concentration of firms in the industry measured by the HHI, there is a negative effect on the probability of closure from schools, which although it has the expected sign, is not statistically significant. For the growth of the school-age population a positive effect is obtained, contrary to the expected. A potential cause of this may be that the relevant market that tries to capture the communal schooling-age population is not relevant for the schools, which could be corrected by using areas around the establishment that allow quantifying the offer with inter-municipal scopes.

The variable duration defines how many periods the establishment is in the base, and it gives an account of the number of years that it is continuously

observed<sup>22</sup>. The positive sign means that as the years go by, an establishment is more likely to close. This could be a consequence of the linear form imposed on time dependence, however, when testing with different functional forms and more flexible specifications (lowess and splines) the result does not change much. That is, the result of the estimation is that the age of the establishment does not decrease the probability of closure, something consistent with the fact that the schools that have closed are mostly municipal (review the figures 1 and ??) with long trajectories in the education system.

Table 5—: Duration Model, closure probability

| Variables                                     | Coefficiente | Err.Std. |
|-----------------------------------------------|--------------|----------|
| Ln. schooling-age population <sub>(t-1)</sub> | 0.111        | 0.109    |
| Total number of students <sub>(t-1)</sub>     | -0.005       | 0.0009** |
| HHI <sub>(t-1)</sub>                          | -2.657       | 1.957    |
| Unemployment Rate <sub>(t-1)</sub>            | 0.066        | 0.136    |
| Rate GDP growth <sub>(t-1)</sub>              | 0.108        | 0.044*   |
| Duration (years since opening)                | 0.100        | 0.049*   |
| Q2-IVE                                        | -0.052       | 0.252    |
| Q3-IVE                                        | 0.149        | 0.249    |
| Q4-IVE                                        | 0.014        | 0.250    |
| Q5-IVE                                        | 0.260        | 0.268    |
| Math SIMCE at 4th grade                       | -0.005       | 0.002*   |
| Dummy Confessional (=1)                       | -5.238       | 1.005**  |
| Dummy school size (=2)                        | -0.001       | 0.241    |
| Dummy school size (=3)                        | -0.609       | 0.272*   |
| Dummy school size (=4)                        | -0.828       | 0.306**  |
| Dummy school size (=5)                        | 0.019        | 0.365    |
| Acts (Type C=1)                               | 0.696        | 0.181**  |
| Urban (=1)                                    | 1.604        | 0.212**  |
| Charter School (=1)                           | -0.194       | 0.156    |
| Intercept                                     | -5.830       | 1.813**  |
| Obs.                                          |              |          |
| N                                             |              | 30.484   |

\* p<0.05; \*\* p<0.01

Within the variables of the economic cycle considered, the average annual unemployment rate has the expected sign. Thus, when unemployment increases in the period  $t - 1$ , the probability of closing the schools increases by  $t$ . On the

<sup>22</sup>Some methodological advantages of the model used lie in its ability to handle late entries (truncations from the left) and desertions (censorship on the right).

other hand, the positive sign of the growth of the lagged GDP in a period draws attention. This significant effect suggests that the education sector is sensitive to a previous boom or crisis, but contrary to the expected one. A potential explanation of the phenomenon may be that closures in a typology of administration (municipal for example) are over compensated by income from another administration (charter or private). The religious orientation of an establishment is closely linked to the antiquity of the schools. The majority of these -mainly Catholics- are the oldest schools in the country. This is why it draws attention to the calculated negative sign, evidence of an inverse relationship that decreases the likelihood of closure of confessional self-declared schools vis-à-vis those that claim to be lay or do not declare any guidance. A fact that suggests that between two very similar schools, one secular and one religious, the layman is more likely to close. There is evidence that the visiting records generated in the inspections, particularly the reports with type C or serious results, have a statistically robust relationship with the probability of closure. This relationship being positive when two or more acts of type C are generated. It should be mentioned that what is found is not synonymous with causality, the eventual closing of a school being the reason for the scheduling of inspection visits (but not its resolution). As serious or very serious). Being a potential source of reverse causality. On the other hand, the positive sign registered by the urban area dummy is in line with the description made in the stylized facts, where it was indicated that the majority of turnover occurs within the urban area. Finally, we have that the number of students negatively affects the probability of closing an establishment, which is directly related to the optimal scale of utilization of installed capacity, captured by school size dummies. When an establishment has very few students operating on a sub-optimal scale, it can become financially untenable to remain open for a longer period.

#### B. Differences in academic results (SIMCE)

For the following model, each student  $i$  is part of the school  $j$ , and there is a hierarchy in two levels (students and schools), with specific random intercepts of each unit  $j$ . The first level error is distributed  $N(0, \sigma_\epsilon^2)$ , while the second level error is distributed  $N(0, \sigma_{u_0}^2)$ :

$$(3) \quad Y_{ij} = \beta_{0j} + \beta_{1j}X_{1ij} + \epsilon_{ij}$$

$$(4) \quad \beta_{0j} = \gamma_{00} + \gamma_{01}Z_j + u_{0j}$$

Replacing the algebraic notation with the model variables, the previous equa-

tions are as follows:

$$\begin{aligned}
 (5) \quad \text{simce}_{ij} = & \beta_{0j} + \beta_1 \ln y_{ij} + \beta_2 \text{mother}_y \text{of } s_{ij} + \beta_3 \text{father}_y \text{of } s_{ij} \\
 & + \beta_4 \text{parents}_{exp} \text{of } i_{ij} + \beta_5 \text{nhousehold}_{ij} + \beta_6 \text{nbooks}_{ij} \\
 & + \beta_7 \text{preschool}_{ij} + \beta_8 \text{repetition}_{ij} + \beta_9 \text{sex}_{ij} + \epsilon_{ij}
 \end{aligned}$$

$$\begin{aligned}
 (6) \quad \beta_{0j} = & \gamma_{00} + \gamma_{01} \text{schsize}_j + \gamma_{02} \text{selection}_j \\
 & + \gamma_{03} \text{confessional}_j + \gamma_{04} \text{urban}_j + \gamma_{05} \text{hhi}_j \\
 & + \gamma_{06} q45_j + \gamma_{07} \text{type}_j + \gamma_{08} \text{closure}_j + u_{0j}
 \end{aligned}$$

The interpretation of fixed effect coefficients is the usual one in a multiple regression model. Noting that for the multilevel case, the coefficients  $\beta_{0j}$  are the intercepts for each establishment  $j$ , incorporating the heterogeneity stemming from the intra-group correlation (within the schools) that reaches 27% this year, of the total variance of the test, a 27% is explained by the variance within each establishment. The most relevant results of the model show that the differences are not statistically significant in favor of the schools that remain open over the schools that close in the results of the SIMCE test of basic rooms. This difference is approximately 5 to 6 points, equivalent to 10-11% of the standard deviation of the test, but with large standard errors in all generated specifications. The differences found support the figure shown in the figure 16, where it is possible to appreciate that they are not uniform in the school population for different levels of the School Vulnerability Index (IVE).

The evidence indicates that there are statistically significant differences in the SIMCE score between those that leave the market and those that do not, particularly for the socioeconomic quintiles 4 and 5 quintiles (situation that can be seen in specification 2 of the table 6). The socio-economic control variables incorporated in the model with the expected positive sign (income and schooling), as well as other controls such as the number of books in the home and attendance at pre-school education. The number of people in the home, the existence of repetition events and the selection of students by the establishment have a negative impact on the score. In addition, there is evidence that students who are part of exchanges in communes with a greater concentration in the market. In most of the tested specifications, the variable that captures the religious orientation of the establishment (equal to 1 if it is confessional), has a positive sign, but not statistically different from zero. On the other hand, there is evidence of a difference not statistically significant between charter schools and municipalities between 1 and 4 points in the SIMCE test, equivalent to approximately 3-6% of the standard deviation of the test, result similar to that found by Lara et al. (2011).

Table 6—: HLM results

| Variables                         | Model 1  |            | Model 2  |            |
|-----------------------------------|----------|------------|----------|------------|
|                                   | Coef.    | Error Std. | Coef.    | Error Std. |
| Log income                        | 1,498    | 0,275**    | 1,497    | 0,275**    |
| Father years of schooling         | 0,6979   | 0,0745**   | 0,6980   | 0,0745**   |
| Mother years of schooling         | 0,9366   | 0,0707**   | 0,9367   | 0,0706**   |
| Parents education expectations    | 6,4130   | 0,1223**   | 6,4130   | 0,1223**   |
| Size of household (total number)  | -0,9095  | 0,0781**   | -0,9093  | 0,0780**   |
| Books in household (total number) | 2,7728   | 0,1459**   | 2,7726   | 0,1464**   |
| Dummy Preschool (assisted=1)      | 1,9408   | 0,8603*    | 1,9672   | 0,8614*    |
| Dummy grade repetition            | -25,1641 | 0,3281**   | -25,1652 | 0,3282**   |
| Sex (Male=1)                      | 4,8599   | 0,4929**   | 4,8605   | 0,4930**   |
| Schools size (categorical)        | 2,5932   | 0,3659**   | 2,5880   | 0,3598**   |
| Dummy of selection in admission   | -0,7712  | 0,1885**   | -0,7733  | 0,1884**   |
| Dummy of type (Confessional=1)    | 0,0574   | 0,5777     | 0,0683   | 0,5727     |
| HHI                               | 14,3975  | 4,4232**   | 14,4923  | 4,3951**   |
| Urban (=1)                        | -7,1355  | 2,1553**   | -7,1103  | 2,1547**   |
| Q4 and Q5 (Not vulnerable=1)      | 11,1759  | 1,2522**   | 11,4233  | 1,2412**   |
| Charter School                    | 1,4315   | 2,5556     | 1,4142   | 2,5502     |
| Private                           | 16,2210  | 1,6103**   | 16,3876  | 1,5779**   |
| Dummy Closure                     | -5,0907  | 1,2147*    | -3,4558  | 1,3639*    |
| Q4 and Q5 * Dummy Closure         |          |            | -7,9979  | 2,0121**   |
| Random Effects                    |          |            |          |            |
| $\sigma_{\epsilon}^2$ (residuo)   | 2,791    | 0,035**    | 2,791    | 0,036**    |
| $\sigma_{u_0}^2$ (intercepto)     | 3,820    | 0,002**    | 3,820    | 0,002**    |
| Obs.                              |          |            |          |            |
| N                                 | 173.751  |            | 173.751  |            |
| Grupos                            | 6.410    |            | 6.410    |            |

\* p&lt;0.05; \*\* p&lt;0.01

*C. Evolution of enrollment by establishment*

The estimated model follows the logic of the equations previously defined but at different levels, for this case level 1 is temporary and 2 corresponds to schools. It also incorporates a random slope for each establishment in addition to the intercept, which makes the notation of the model a bit more complex, which in general terms is expressed following Hox (2002) in the following way:

$$Y_{ij} = \gamma_{00} + \sum_p \gamma_{p0} X_{pij} + \sum_q \gamma_{0q} Z_{qj} + \sum_p \sum_q \gamma_{pq} X_{pij} Z_{qj} + \sum_p u_{pj} X_{pij} + u_{0j} + \epsilon_{ij}$$

The vector of first level variables is simply the year, that is, the vector dimension  $X$  is  $p = 1$ . The dimension of the vector  $Z$  containing the regressors at the level of schools is  $q = 11$ : the SIMCE mathematics score of 4th grade, the concentration HHI of the industry at the municipal level, the socioeconomic group of the establishment, the repetition measure as the percentage of students who have repeated the total number of students in the establishment, preschool education measured as the percentage of students who made preschool education of the total, the logarithm of the school-age population of the commune, the generation of records of visits type C (as a dichotomous variable equal to 1 when more than two minutes are generated per year), a dichotomous variable of geographical area and administrative dependency (equal to 1 when it is private, or charter), first lag of the logarithm of the number of students of the establishment and finally the trend (year). It is found that the registration of the private schools is strongly affected by the results of the SIMCE test, particularly significant is the effect found for the SIMCE of the fourth grade in the Metropolitan Region. Registering a statistically significant positive sign (see table 7), reflecting that on average the schools with the best SIMCE scores (after controlling for socioeconomic, demographic and other factors) have more students<sup>23</sup>. A positive relationship is observed between the industry concentration index (HHI) and the number of enrolled students, a situation that is statistically significant at 1%. In this sense, the interpretation is direct, since, at higher levels of market concentration, schools on average have a greater number of students.

There is a negative effect of the variable inspection visits minutes, and although it is not statistically significant in the final specification, it is evidence of a trend (see annexes) relevant to the dynamic implications that a malfunction evidenced by the record-keeping C, may have in the decision of the attorneys to keep students in the establishment. From the model it is possible to infer that, on average, a school that presents more than two inspections that give rise to C-type certificates in a year (when the violations are considered to be serious.<sup>24</sup>), has a lower number of students enrolled. The controls of rurality and school-age population show facts known as: that rural schools have fewer students on average than their urban peers, and that compared to a growth in local demand (population of

<sup>23</sup>This effect found is robust to the functional specification and regressor used, changing the SIMCE in level by percentage differences with respect to the average communal SIMCE, standard deviations or a 'quintilization' of the SIMCE at the national level, the result is maintained (review annexes for more details).

<sup>24</sup>Examples of serious infractions are: 1) not having a certificate of environmental hygiene or not having it updated, 2) not having a certificate of definitive reception of the school extended by the Municipal Works Directorate, 3) not having fire extinguishers, and 4) infractions associated with registration of registration and attendance of students that affect the payments of grants.



Table 7—: Multilevel Longitudinal Model

| Fixed effects                        | Coef.   | Err.Std.  |
|--------------------------------------|---------|-----------|
| Math SIMCE <sub>(t-1)</sub>          | 0.003   | 0.00**    |
| HHI                                  | 0.15    | 0.05**    |
| Socioeconomic Group                  | 0.01    | 0.00**    |
| Grade repetition (rate)              | 0.03    | 0.01**    |
| Students with preschool (rate)       | 0.05    | 0.01**    |
| Ln. schooling-age population         | 0.00    | 0.00*     |
| Acts (Type C=1)                      | -0.01   | 0.00      |
| Urban (=1)                           | 0.02    | 0.01**    |
| Private School (=1)                  | -0.04   | 0.01**    |
| Log. Student Number <sub>(t-1)</sub> | 0.98    | 0.00**    |
| Year                                 | -0.01   | 0.00**    |
| Random Effects                       |         |           |
| $\sigma_{\epsilon}^2$ (residual)     | 0.01    | 0.00**    |
| $\sigma_{u_0}^2$ (intercept)         | 0.001   | 0.00**    |
| $\sigma_{u_1}^2$ (slope=year)        | 2.1e-15 | 1.7e-15** |
| Obs                                  |         |           |
| N                                    | 14.704  |           |
| Groups                               | 2.640   |           |

\* p&lt;0.05; \*\* p&lt;0.01

school age) the number of students in schools obviously increases. A third control variable is time, where the downward trend of the total enrollment in recent years is captured by the negative and significant slope of the year variable. On the other hand, schools with a higher number of repeating students have statistically more students on average than other schools, which may be evidence of high rates of incorporation and/or low dropout (high retention). This can be due to many reasons, among them a high capture of the students by uncompetitive offer levels, or the incentives not to skim the repeating students for the received school subsidy. Finally, there is a significant relationship with the proportion of vulnerable students enrolled, with a positive and significant sign at 1%. That is to say, increases in the vulnerability of the schools generate drops in their enrollment, particularly in the municipal schools. This last finding is not surprising, and is aligned with the evidence of a constant decline in municipal enrollment since 1990 (at an annual rate close to 1%).

#### D. Total Turnover (Communes)

The model used follows the previous equations, where each commune  $i$  is part of the region  $j$ , and as in the previous model, it has a hierarchy in two levels, with specific random intercepts of each unit  $j$ . The first level error is distributed  $N(0, \sigma_\epsilon^2)$ , while the second level error is distributed  $N(0, \sigma_{u_0}^2)$ . The estimates confirm the previously described showing that the total turnover has a significant and positive relationship with the level of concentration of the industry, while the higher the Herfindahl-Hirshmann index (few schools agglomerate high demand) generates more inputs and outputs of schools, particularly in those communes where the proportion of charter schools is low, and the presence of municipal schools is high. Both the aggregate vulnerability of the commune and the performance in SIMCE relative to national performance, have significant effects on the closure of schools. For the former, the relationship is negative, that is, in communes in the least vulnerable quintiles there are fewer closures from schools compared to the most vulnerable communes. The opposite occurs with SIMCE, although this result is less robust to the specification (review annexes), where in communes with higher SIMCE results there were more closures on average in the 20 years analyzed than in communes with lower results in test.

Table 8—: Hierarchical Linear Model, Closures

| Fixed Effects                  | Coef.  | Error Std. |
|--------------------------------|--------|------------|
| HHI                            | 0.524  | 0.14**     |
| Ln. Municipal Schools          | 0.184  | 0.03**     |
| Ln. schooling-age population   | 0.094  | 0.02**     |
| Average SIMCE                  | 0.002  | 0.0008**   |
| Entry <sub>(t-1)</sub>         | 0.034  | 0.02       |
| Entry                          | 0.056  | 0.02**     |
| Closure <sub>(t-1)</sub>       | 0.133  | 0.02**     |
| Socioeconomic Group            | -0.032 | 0.01*      |
| Urban (=1)                     | 0.050  | 0.04       |
| Year                           | 0.012  | 0.00**     |
| Random effects                 |        |            |
| $\sigma_\epsilon^2$ (residual) | 0.568  | 0.013**    |
| $\sigma_{u_0}^2$ (intercept)   | 0.008  | 0.004**    |
| Obs                            |        |            |
| N                              |        | 3.414      |
| Groups                         |        | 15         |

\* p<0.1; \*\* p<0.01

There are significant dynamic effects captured by the lagged output variables one period and the variable of total municipal inputs. This reflects a certain tendency or inertia in the first case, and presumably effects of displacement of one dependency by another in the second. In particular, the model provides evidence that supports the thesis that in communes where a large number of entries are produced in a year, a large number of closures will take place in that same year. The evidence suggests that rurality does not play a significant role in this relationship.

#### IV. Final Thoughts

Evidence is found that supports the following conclusions: The schools that close, regardless of their administrative dependence, on average have more vulnerable students. Second, the schools that close have lower performance in any of the standardized performance tests (SIMCE and PSU), but when correcting for vulnerability level of the school the evidence is mixed, and only those out of the least vulnerable quintiles have statistically more results low than incumbents that do not close. Third, the evidence suggests that the inspection and inspection mechanisms carried out by the state have effects on the probability of closure of an establishment, especially when type C acts are generated. This result, however, does not imply causality, so it must be taken with caution. Suggesting that as the binding role of oversight and infractions is strengthened, transversal improvements of the Shumpeterian type in the education system will effectively be generated. Fourth, schools that close present average monthly collections (shared financing) significantly lower than those that do not close, and major differences with respect to those that enter, phenomenon could be explained simply by the changes in the composition of administrative dependencies towards a with a higher prevalence of charter students subject to shared financing. Fifth, the concentration of the industry has important effects on total turnover, showing that the scale of the relevant market is key, and the areas of incidence and its growth (school-age population) will strongly determine the total turnover that is generated, arising with This is a challenge in methodologically defining these areas for further analysis. Finally, it should be noted that the closure of the schools can have undesirable effects in many aspects. In particular, there are relocation costs internalized by families that seek to give continuity to the educational process of the students and learning costs for the students. This work suggests that it mainly affects the most vulnerable in the country.

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## APPENDIX

### A1. Statistical robustness analysis

The selected models are *stressed*, to verify if the results are robust, that is to say, stable against changes in the vector of regressors, methodology of estimation of the parameters, or data sample occupied. Figure A1 shows a summary of the confidence intervals and coefficients for different specifications of the linear hierarchical model, with no significant changes in the presented conclusions. The first two models are simple OLS, the first with identical vector of regressors that the final model occupied (HLM\*), the second does not have the controls at the school or municipal level, both incorporate standard errors grouped by school (clustered errors). The fourth model is an HLM with the SIMCE test of language as a dependent variable. The fifth and sixth models are HLM with intercepts and random slopes, and standard errors grouped by region and school respectively.

An additional comparison exercise is carried out, in which medians are imputed by establishing some control variables at the student level (schooling and income) to obtain a lower attrition in them. The final sample has 90% of the total records, vis-a-vis the original 72%, and the results support the conclusions suggested in the paper.

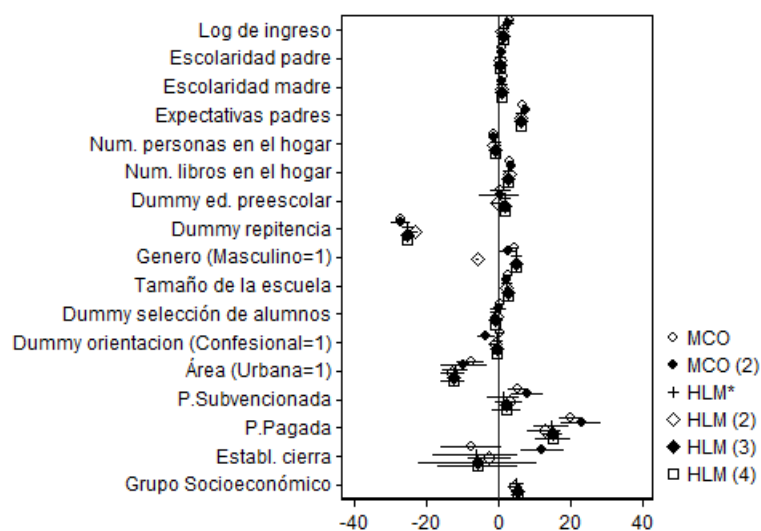


Figure A1. : Hierarchical Linear Model, Coefficients and confidence intervals

Source: Based on MINEDUC data.

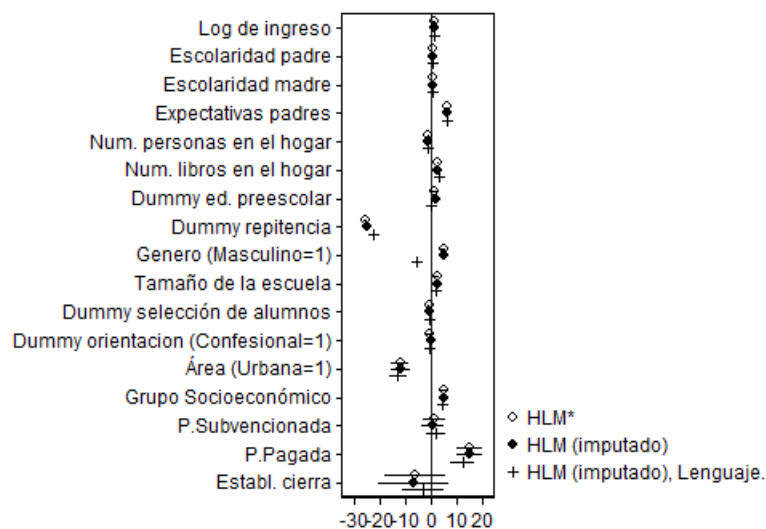


Figure A2. : RHierarchical Linear Model (original and with imputations)

Source: Based on MINEDUC data.